

Report Template (Underactuated Robotics)

Your Name

1 Introduction

Introduce briefly the addressed modeling and control problem.

2 System Model

Insert a diagram of the considered system and description of all variables and parameters.

2.1 Euler-Lagrange Modeling

Use Euler-Lagrange Modeling to derive the equations of motion for the considered system. Put details in appendix

2.2 Analysis of Model

Linearize the model and determine if it is controllable.

3 Controller Design

Use two different nonlinear control methods for controlling the system. Remember to insert references, e.g. [1]. You should include some analysis of stability.

3.1 Controller Design Method 1

3.2 Controller Design Method 2

4 Simulation Results

Insert graphs from numerical simulations. Include both control signal u and state x or output y . It would also be good to show the total mechanical energy and work done by friction/actuation

5 Conclusions

Conclude briefly on the results and project.

References

- [1] R. Xu and Ümit Özgüner, “Sliding mode control of a class of underactuated systems,” *Automatica*, vol. 44, no. 1, pp. 233–241, 2008.

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